

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++, SQL, Javascript, TypeScript

Deep Learning: PyTorch, TensorFlow, Deep Learning Models, Medical Imaging, 3D Segmentation, Computational Geometry

MLOps & Cloud: Docker, Kubernetes, AWS SageMaker, FastAPI, AWS-based Cloud Products, Real-time System Optimization

Data Engineering: PySpark, Airflow, ETL Pipelines, Data Processing

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and rigorously evaluated **deep learning models** for metrological applications, enhancing **model performance, training efficiency, and numerical stability**.
- Explored novel **DL architectures** and physics-informed hybrid loss functions to significantly improve **algorithm accuracy and robustness**.
- Collaborated effectively with PhD researchers and cross-functional teams, contributing to research and a publication for new methods.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Optimized and deployed **3D segmentation algorithms** for **medical image analysis** (e.g., vitiligo detection), achieving 94% accuracy and 5x speed improvement using Docker and FastAPI.
- Engineered a real-time **deep learning** system for tennis foul detection with TensorRT, demonstrating 6x reduction in computational cost and time, emphasizing **algorithm optimization and efficiency**.
- Managed robust **MLOps pipelines on AWS SageMaker**, including addressing data drift for YOLO models and maintaining Kubernetes infrastructure.
- Developed and validated proof-of-concept **deep learning models** (YOLO, few-shot learning, CNN) for client demonstrations, translating clinical/business needs into robust **algorithm designs**.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Led full-stack development of AI-based web services using **Python** (Flask), contributing to scalable platform architecture.

- Developed an AI chatbot application leveraging the OpenAI API and LLM frameworks, demonstrating rapid prototyping of **deep learning**-driven features.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Designed and implemented robust ETL pipelines for structured and unstructured data, ensuring high-quality data ingestion for analytical and **model training workflows**.
- Automated complex data cleaning and ingestion workflows, improving data **accuracy** and **efficiency** for diverse client analytics use cases.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected and validated an ensemble of **deep learning models** (DPFunc, DeepGOLPlus) for protein function prediction, leveraging AlphaFold for advanced feature extraction.
- Achieved a 4% increase in **model accuracy** through custom feature engineering and **rigorous validation**, demonstrating strong **algorithm design** capabilities.

Razanalytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed and validated **deep learning models** for **real-time** Bitcoin price forecasting, emphasizing **numerical stability** and robust prediction **algorithms**.
- Optimized model architectures and data processing pipelines for **computational efficiency** and **real-time** inference in a high-frequency data environment.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Designed a **real-time deep learning** system for music generation using transformer-based models, integrating complex signal processing for temporal dependencies.
- Implemented and optimized **algorithms** for sequence prediction in audio generation, demonstrating strong understanding of neural network architectures and **computational efficiency**.

Hugging Face

Open-Source

Contributor

- Co-authored technical documentation and implementation for Zero-shot Image Classification, showcasing expertise in **deep learning models** and clear communication of complex **algorithms**.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized in **deep learning** for computer vision and spatio-temporal modeling
- Relevant coursework: Advanced Neural Networks, Computer Vision, Autonomous Systems

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor
- Specialized in Machine Learning and Computer Vision with focus on time-series analysis

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python